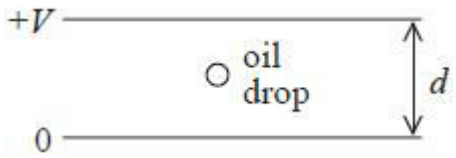
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Questions

Q1.

Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .

An oil drop with charge *q* and mass *m* remains stationary between two metal plates with a potential difference *V* across them. The plates are separated by distance *d*.



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What is the charge on the oil drop?

☐ **A** $\frac{md}{V}$

☐ **B** $\frac{mV}{d}$

☐ **C** $\frac{mgd}{V}$

☐ **D** $\frac{mgV}{d}$

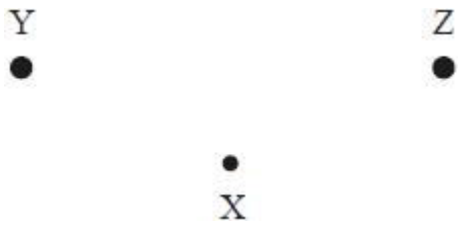
(Total for question = 1 mark)

Q2.

Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .

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Two equal positive charges are placed at Y and Z.

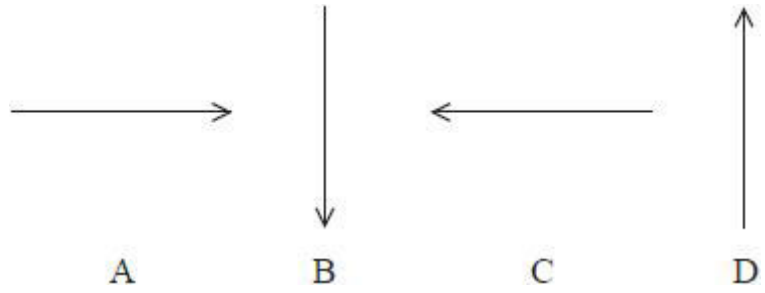


Point X is equidistant from Y and Z.

Which of the following shows the direction of the electric field at point X?

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☐ **A**

☐ **B**

☐ **C**

☐ **D**

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(Total for question = 1 mark)

Q3.

Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .

When a point object with charge +Q is placed a distance d away from another point object with charge +2Q the force between them is F.

The charges are increased to +2Q and +4Q and the distance is increased to 2d.

Which of the following gives the new force?

☐ **A** $\frac{F}{4}$

☐ **B** $\frac{F}{2}$

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☐ **C** F

☐ **D** $2F$

(Total for question = 1 mark)

Q4.

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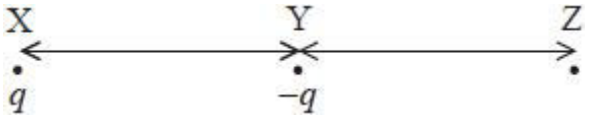
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Point charges of $+q$ and $-q$ are placed at X and Y respectively.

When another charge is placed at Z, the resultant force on the charge at X is zero.

Distance XY equals distance YZ.

Which of the following is the charge placed at Z?



☒ **A** $\frac{q}{4}$

☒ **B** $\frac{q}{2}$

☒ **C** $2q$

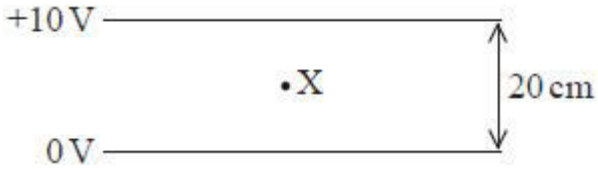
☒ **D** $4q$

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(Total for question = 1 mark)

Q5.

The diagram shows a potential difference applied across two conducting parallel plates.



Point X is midway between the plates.

Which of the following is the electric field strength at point X?

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☒ **A** 50 V m^{-1} upwards

☒ **B** 50 V m^{-1} downwards

☒ **C** 100 V m^{-1} upwards

☒ **D** 100 V m^{-1} downwards

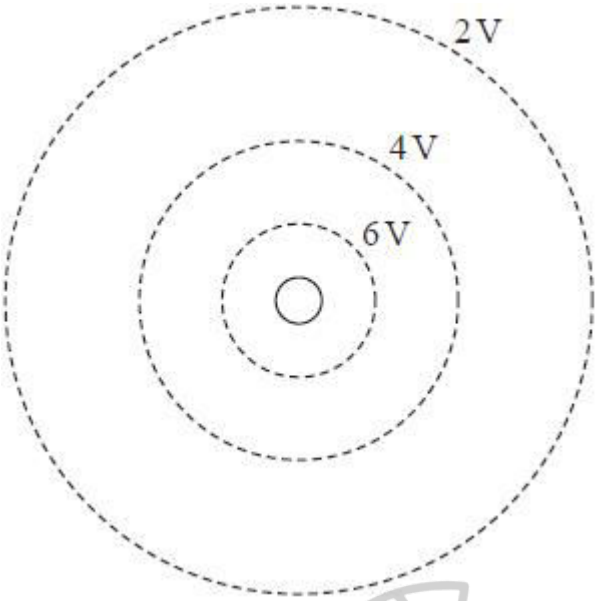
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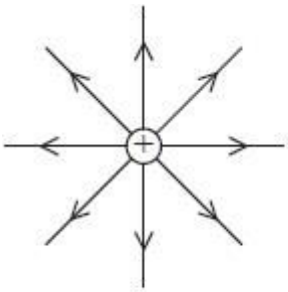
Q6.

The diagram shows equipotential lines about a point charge.

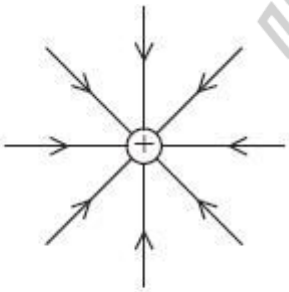


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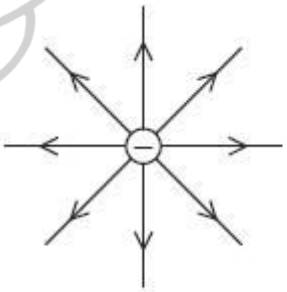
Which of the following diagrams shows a possible combination of electric field lines and polarity of point charge to produce this pattern?



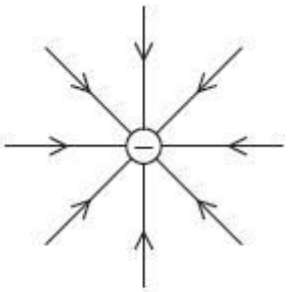
☒ **A**



☒ **B**



☒ **C**



☒ **D**

(Total for question = 1 mark)

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Q7.

At distance d from a point charge of magnitude q the potential is V.

Which of the following is the potential at a distance $4d$ from a point charge of magnitude $2q$?

☒ **A** $\frac{V}{4}$

☒ **B** $\frac{V}{2}$

☒ **C** V

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☒ **D** $2V$

(Total for question = 1 mark)

Q8.

Some spiders may be seen suddenly moving up into the air, even when there is no wind. This happens when they let out strands of web.

The spiders are pulled upwards as the electrically charged strands of web interact with the Earth's electric field.

The average strength of the Earth's electric field is 120 V m^{-1} downwards.

(a) State, with a reason, the polarity of the charge on the strands of web.

(1)

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(b) A spider moves upwards after letting out a strand of web.

Determine the initial upwards acceleration of the spider.
mass of spider = 3.00×10^{-6} kg
total magnitude of charge on a strand of web = 3.00×10^{-7} C

(5)

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Acceleration = ...

(c) The Earth has an electric field because charge is distributed over its surface.

Determine the quantity of charge on 1 m^2 of the Earth's surface which would cause an electric field of

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120 V m^{-1}.
You should assume that the charge is distributed evenly.
radius of Earth = 6400 km

(3)

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Charge = ...

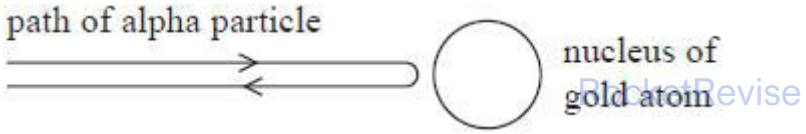
(Total for question = 9 marks)

Q9.

In the early part of the 20th century, experiments were carried out in which alpha particles were directed at thin sheets of metal.

A few alpha particles were deviated through small angles and a very small proportion were reflected back.

(a) The diagram represents an alpha particle reflected back through 180° as it approached the nucleus of a gold atom.



Calculate the maximum acceleration of the alpha particle as it reaches the point of minimum separation from the nucleus. Assume that the gold nucleus remains at rest.
speed of alpha particle = 1.74×10^{7} m s^{-1}
mass of alpha particle = 6.64×10^{-27} kg
atomic number of gold = 79

(5)

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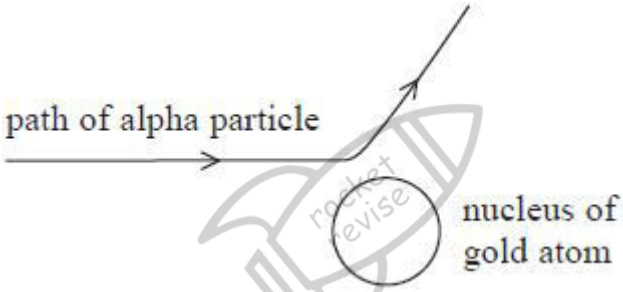
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Maximum acceleration = ...

(b) The diagram represents the path of an alpha particle with the same initial speed as in (a) that is deflected through a smaller angle.



Explain whether the maximum acceleration would be the same as for the alpha particle reflected back through 180°.

(4)

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(Total for question = 9 marks)

Q10.

In 1909 Robert Millikan carried out experiments to determine the magnitude of the charge on an electron.

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His experiments involved the motion of small, electrically charged droplets of oil between two charged metal plates.

Some students carried out a similar experiment in a school laboratory.

(a) The terminal velocity v of an oil droplet was measured as it fell a known distance in air when the plates were uncharged. Stokes' law was then used to determine the radius r of the oil droplet. Upthrust was ignored.

(i) Show that $r = \sqrt{\frac{9\eta v}{2\rho g}}$

η = viscosity of air
ρ = density of oil

(3)

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(ii) For a particular oil droplet, the terminal velocity was measured to be 5.35×10^{-4} m s^{-1}. Calculate the radius of this oil droplet.
$\eta = 1.86 \times 10^{-5}$ Pa s
ρ = 904 kg m^{-3}

(2)

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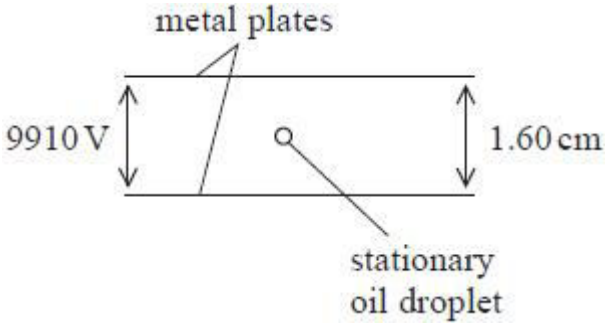
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Radius = ...

(iii) A potential difference (p.d.) was applied across the plates and adjusted until a charged oil droplet was stationary between them.

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Calculate the charge on the oil droplet.
mass of oil droplet = 3.03×10^{-14} kg
p.d. = 9910 V

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separation of plates = 1.60 cm

(4)

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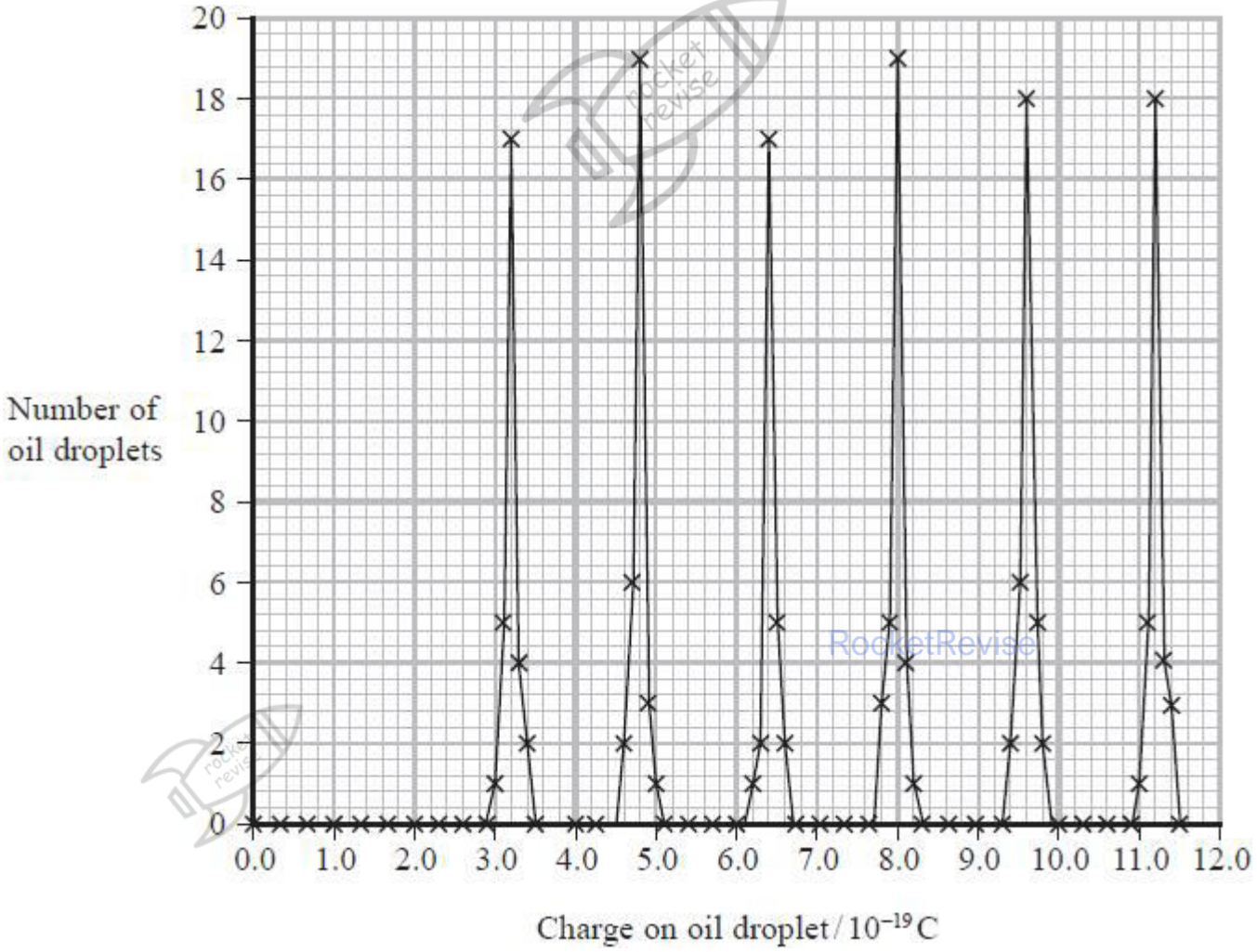
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Charge = ...

(b) The students repeated the experiment for a large number of oil droplets.

The graph shows the number of oil droplets with each measured charge.



The teacher states that charge can only be transfered in integer multiples of 1.6×10^{-19} C.
Explain the extent to which the students' results support this statement.

(3)

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(c) In 1909 Millikan concluded that the charge on the electron was 1.592×10^{-19} C whereas the accepted value today is 1.602×10^{-19} C. This was because the value for the viscosity of air used by Millikan was incorrect.

Deduce whether the value for the viscosity used by Millikan was too large or too small.

(3)

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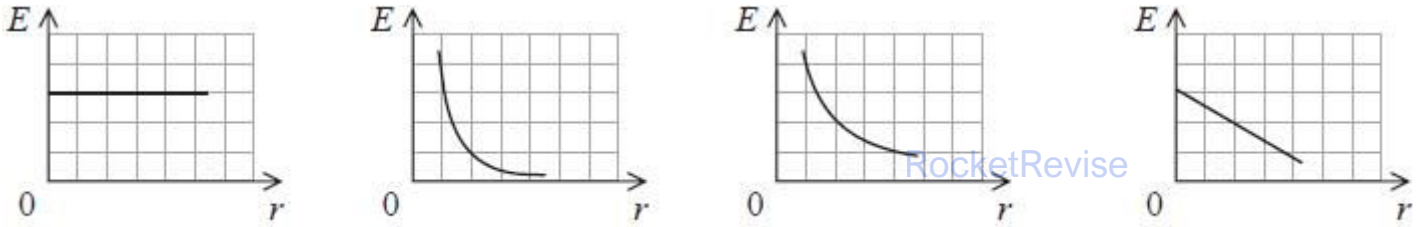
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(Total for question = 15 marks)

Q11.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Which of the following graphs shows how the electric field strength *E* varies with distance *r* from a positive point charge?



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☒ **A**
☒ **B**
☒ **C**
☒ **D**

(Total for question = 1 mark)

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Q12.

Point charges of 2 μC and 3 μC are placed 1.2 m apart as shown.



What is the electric field strength at the point labelled X?

☐ **A** $\frac{2\times10^{-6}}{4\pi\varepsilon_0(0.4)^2}+\frac{3\times10^{-6}}{4\pi\varepsilon_0(0.8)^2}$

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☐ **B** $\frac{2\times10^{-6}}{4\pi\varepsilon_0(0.4)^2}-\frac{3\times10^{-6}}{4\pi\varepsilon_0(0.8)^2}$

☐ **C** $\frac{2\times10^{-6}}{4\pi\varepsilon_0(0.8)^2}+\frac{3\times10^{-6}}{4\pi\varepsilon_0(0.4)^2}$

☐ **D** $\frac{2\times10^{-6}}{4\pi\varepsilon_0(0.8)^2}-\frac{3\times10^{-6}}{4\pi\varepsilon_0(0.4)^2}$

(Total for question = 1 mark)

Q13.

A 6 V battery is connected across two parallel metal plates.

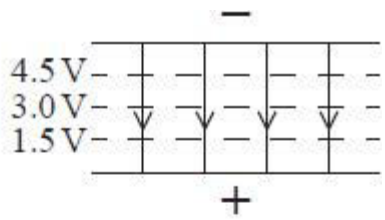
Which of the following shows the electric field lines and equipotentials in the region between the plates?

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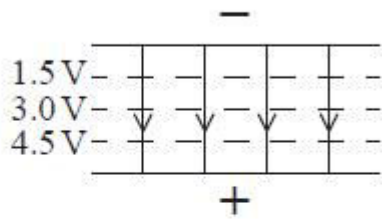
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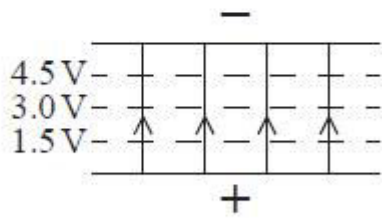
☒ **A**



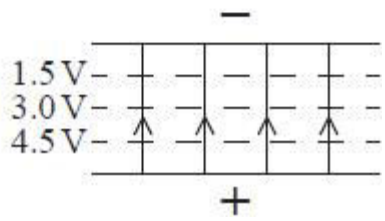
☒ **B**



☒ **C**



☒ **D**



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(Total for question = 1 mark)

Q14.

In 1931, Sloan and Lawrence built a linear accelerator (linac) with several drift tubes. They used the linac to accelerate mercury ions up to energies of 1.26 MeV. The behaviour of the particles was non-relativistic.

(a) The kinetic energy of a non-relativistic particle of mass m with momentum p is given by

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$$E_k = \frac{p^2}{2m}$$

Derive this formula.

(2)

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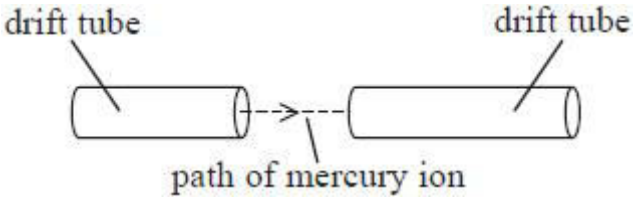
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(b) A mercury ion with kinetic energy 6.42×10^{-15} J leaves a drift tube, as shown.

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Calculate the momentum of the mercury ion when it reaches the next drift tube.
mass of mercury ion = 3.32×10^{-25} kg
charge of mercury ion = 1.60×10^{-19} C
electric field strength between drift tubes = 7.64×10^{6} V m^{-1}
distance between drift tubes = 5.50×10^{-3} m

(4)

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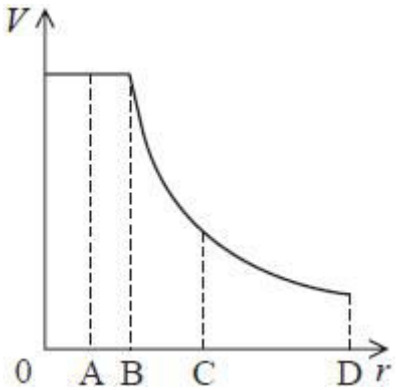
Momentum = ...

(Total for question = 6 marks)

Q15.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

The electric potential V varies with the distance r from a charged object as shown.



At which distance, A, B, C or D, does the electric field strength have its maximum value?

☒ **A**

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☒ **B**
☒ **C**
☒ **D**

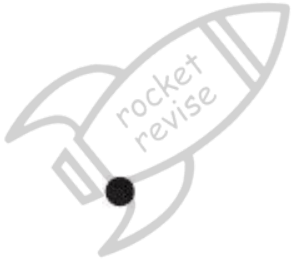
(Total for question = 1 mark)

Q16.

The diagram represents a negative point charge.

Draw field lines to show the electric field around the charge.

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(Total for question = 3 marks)

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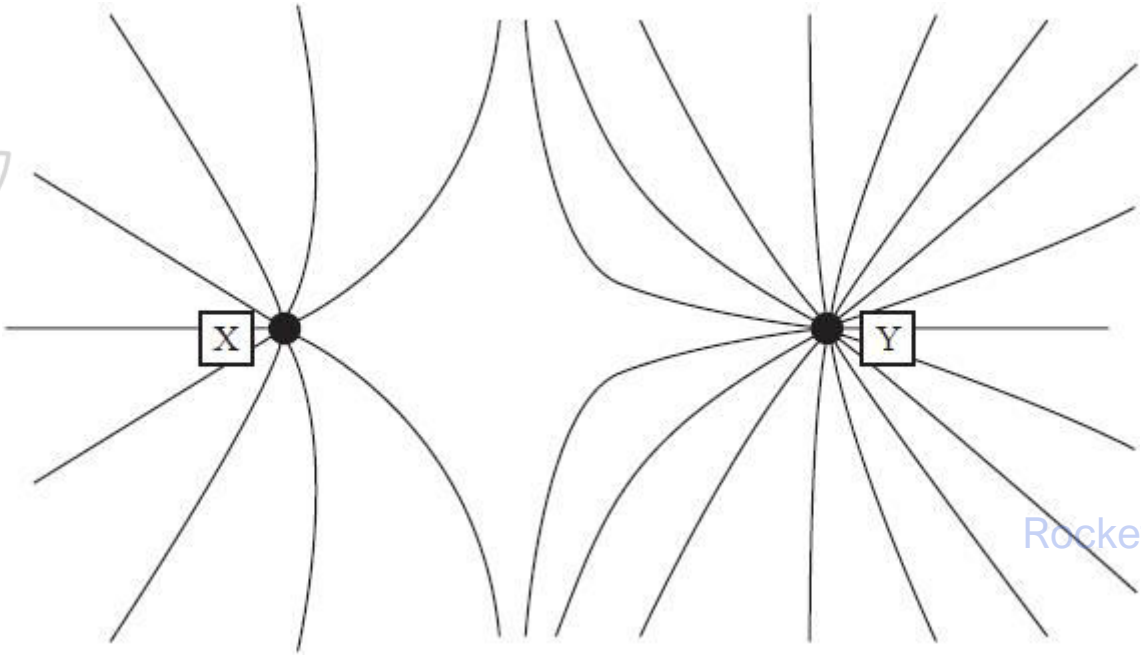
Q17.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

The diagram shows the shape of the electric field pattern between two charged particles, X and Y.

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X has a charge of –*Q*.

Which of the following could be the charge of Y?

- ☒ **A** –*Q*
- ☒ **B** –2*Q*
- ☒ **C** +*Q*
- ☒ **D** +2*Q*

(Total for question = 1 mark)

Q18.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Two equally-charged ions exert a force of 4.0×10^{-9}N on each other. The distance between the centres of the ions is 2.4×10^{-10}m.

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Which of the following is equal to the charge on each ion?

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☒ **A** $\sqrt{\dfrac{4.0\times10^{-9}\times\left(2.4\times10^{-10}\right)^2}{8.99\times10^{9}}}$

☒ **B** $\sqrt{\dfrac{4.0\times10^{-9}\times\left(2.4\times10^{-10}\right)^2}{4\times\pi\times8.85\times10^{-12}}}$

☒ **C** $\dfrac{4.0\times10^{-9}\times\left(2.4\times10^{-10}\right)^2}{8.99\times10^{9}}$

☒ **D** $\dfrac{4.0\times10^{-9}\times\left(2.4\times10^{-10}\right)^2}{4\times\pi\times8.85\times10^{-12}}$

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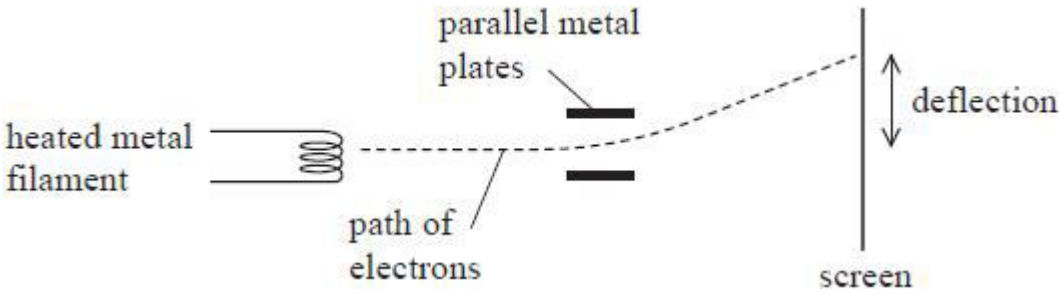
(Total for question = 1 mark)

Q19.

Select one answer from A to D and put a cross in the box ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

This question refers to a cathode ray tube.

The diagram shows some of the components in a cathode ray tube.



When a potential difference is applied across the parallel metal plates, the electrons are deflected.

Which of the following changes would decrease the deflection of the electrons?

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☒ **A** increasing the current in the metal filament
☒ **B** increasing the distance between the plates
☒ **C** increasing the length of the plates
☒ **D** increasing the potential difference across the plates

(Total for question = 1 mark)

Q20.

Answer the question with a cross in the box you think is correct ☒. If you change your mind

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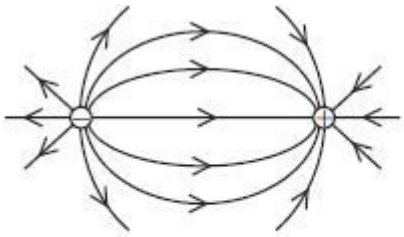
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about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

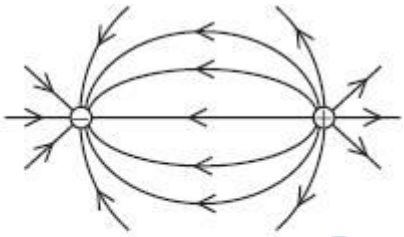
Point charges of equal magnitude are placed close together, as shown.



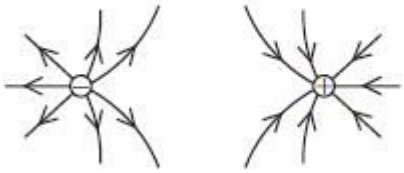
Which of the following diagrams shows the electric field around the charges?



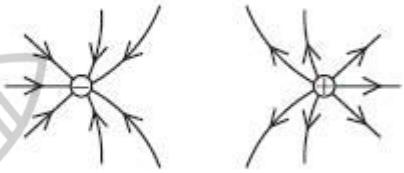
A



B



C



D

- ☒ A
- ☒ B
- ☒ C
- ☒ D

(Total for question = 1 mark)

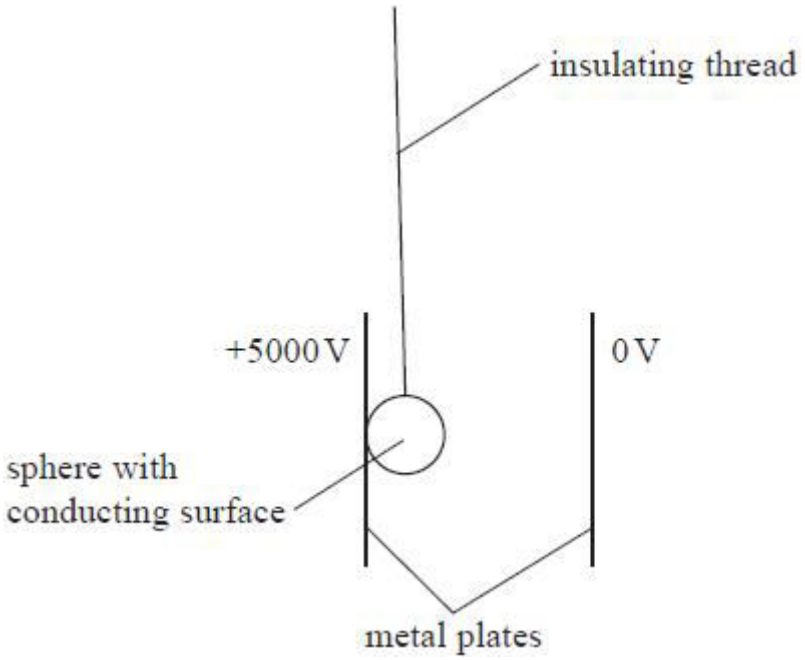
Q21.

A potential difference of 5000 V is applied across two vertical metal plates.

A sphere with a conducting surface is suspended by an insulating thread and touches the positively charged plate as shown.

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The sphere becomes positively charged.

(a) Complete the diagram to show the electric field around a positively charged sphere.

(3)



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(b) (i) Show that the charge on the sphere is about 10 nC.

potential at surface of sphere = 5000 V radius of sphere = 20 mm

(2)

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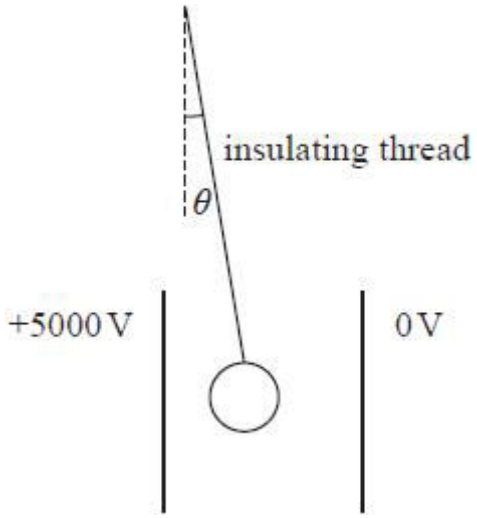
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(ii) The sphere moves away from the positive plate and comes to rest at an angle θ to the vertical.



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Show that the horizontal force on the sphere is about 5×10^{-4} N.
distance between plates = 10.5 cm

(3)

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(iii) Show that θ is about 1°.
mass of sphere = 2.7 g

(3)

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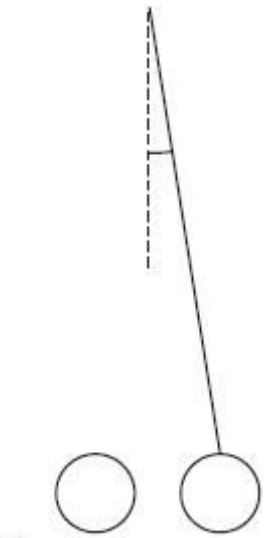
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(c) A second identical charged sphere is held in a fixed position.

The first sphere, attached to the insulating thread, is placed near to the fixed sphere.
The spheres exert a repulsive force on each other.

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The force between the spheres is 5.0×10^{-4} N.
Calculate the distance between the centres of the spheres.
charge on each sphere = 12 nC

(3)

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Distance between centres of spheres = ...

(Total for question = 14 marks)

Q22.

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A point positive charge and a point negative charge are placed 8.0 cm apart at X and Y, as shown.



(a) Calculate the magnitude of the electric field strength midway between X and Y.

charge at X = + 2.5×10^{-7} C
charge at Y = – 2.5×10^{-7} C

(3)

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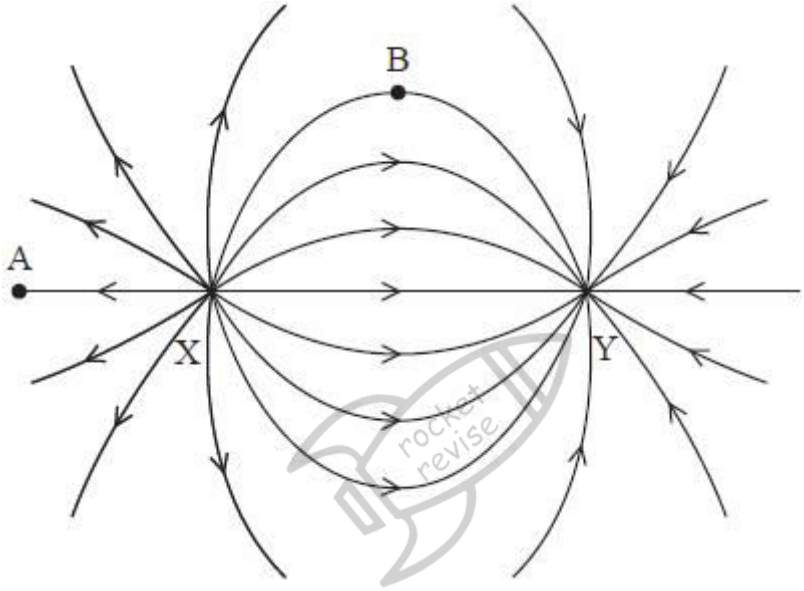
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Electric field strength = ...

(b) The diagram below represents the electric field for this combination of charges.



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(i) Add dashed lines to the electric field diagram to show equipotentials for this combination of charges.

(3)

(ii) A textbook states, "An electric field line shows the path a free positive test charge follows". Discuss the accuracy of this statement for free positive test charges placed at point A and at point B.

(4)

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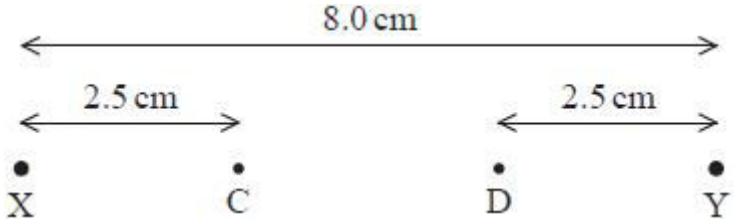
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(c) The charges at X and Y are replaced by charges of twice the magnitude.

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C and D are points between the charges.
Determine the magnitude of the potential difference between points C and D.
charge at X = $+5.0 \times 10^{-7}$ C
charge at Y = -5.0×10^{-7} C

(4)

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Magnitude of potential difference = ...

(Total for question = 14 marks)

Q23.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross and then mark your new answer with a cross ☒.

A proton is accelerated by a uniform electric field. After time *t* the speed of the proton has increased by Δv.

A helium nucleus $^{4}_{2}\mathrm{He}$ is accelerated by the same electric field.

After what time will the helium nucleus achieve the same change in speed?

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☐ **A** $4t$

☐ **B** $2t$

☐ **C** $\frac{t}{2}$

☐ **D** $\frac{t}{4}$

(Total for question = 1 mark)

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Q24.

A point charge is placed at point X, as shown.



(a) Add lines to the diagram to show equipotentials at intervals of equal potential difference.

(2)

(b) A charge of –4.5 nC is placed 4.0 cm from X.

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Calculate the magnitude of the force acting on is charge due to the charge at X.
charge at X = +7.0 nC

(2)

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Magnitude of force = ...

(c) The –4.5 nC charge is moved from a distance of 4.0 cm from X to a distance of 9.0 cm from X.

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Calculate the work done on the -4.5 nC charge.

(3)

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Work done = ...

(Total for question = 7 marks)

Q25.

In the early universe, a helonium molecule formed when a proton joined with a helium atom.

The helium atom in the helonium molecule could lose both of its electrons in a collision.
When this happened, the helium ion and the proton accelerated away from each other.

The initial separation of the helium ion and the proton was 7.9×10^{-11} m.

(a) Calculate the initial force between the helium ion and the proton.

(2)

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Initial force between helium ion and proton = ...

(b) Calculate the electric field strength due to the helium ion, at the initial position of the proton.

(2)

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Electric field strength due to helium ion = ...

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(Total for question = 4 marks)

Q26.

Scientists are developing safe methods for removing old satellites from orbit.

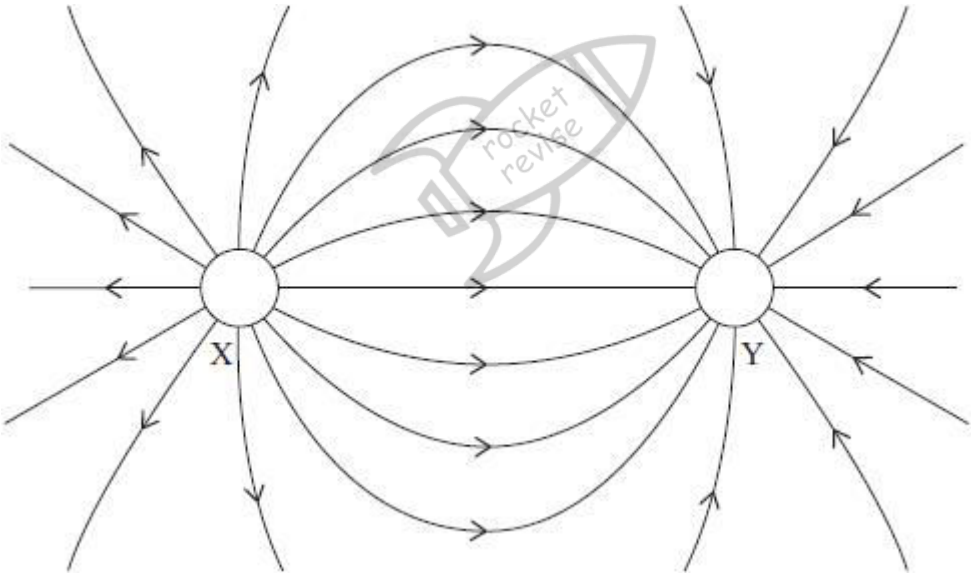
Some scientists plan to use electrostatic forces. A spacecraft will fire a beam of electrons at a satellite. This will give the satellite a negative charge and the spacecraft a positive charge.

There will be an electrostatic force of attraction between the satellite and the spacecraft.

The spacecraft will then move away, taking the satellite with it.

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(a) The diagram shows the electric field around the spacecraft X and the satellite Y when they have equal and opposite charge. You may assume that the spacecraft and the satellite are both spherical and have the same diameter.



(i) Add dashed lines to the electric field diagram to show equipotentials at intervals of equal potential difference.

(3)

(ii) Label the equipotential that represents 0 V.

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(1)

(b) After some time, the charge on the spacecraft will be $+1.5 \times 10^{-6}$ C and the charge on the satellite will be -1.5×10^{-6} C.

(i) Calculate the minimum energy, in joules, required for an electron leaving the surface of the spacecraft to reach the surface of the satellite.

Assume that the charge on each object does **not** affect the potential at the surface of the other object.

radius of satellite = 2.5 m

radius of spacecraft = 2.5 m

(3)

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Minimum energy = ... J

(ii) The spacecraft moves to a new position, taking the satellite with it.
The distance between the satellite and the spacecraft remains constant, so the electrostatic force is constant.
The scientists estimate that the satellite could be moved a distance of 300 km in about 60 days.
Deduce whether this estimate is correct.
Assume gravitational forces are negligible and the satellite is initially at rest.
distance between centre of satellite and centre of spacecraft = 20 m
mass of satellite = 2500 kg

(4)

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(c) In reality, the magnitude of the positive charge on the spacecraft may be greater than the magnitude of the negative charge on the satellite.

Suggest why.

(1)

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(Total for question = 12 marks)

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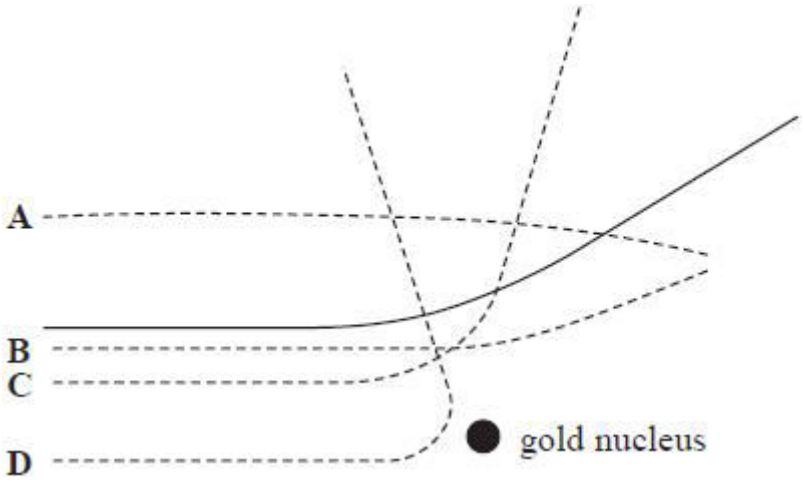
Q27.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

The solid line shows the path of an alpha particle as it passes close to a gold nucleus.

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A second alpha particle with the same kinetic energy approaches the gold nucleus.

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Which dotted line shows a possible path for the second alpha particle?

- ☒ **A**
- ☒ **B**
- ☒ **C**
- ☒ **D**

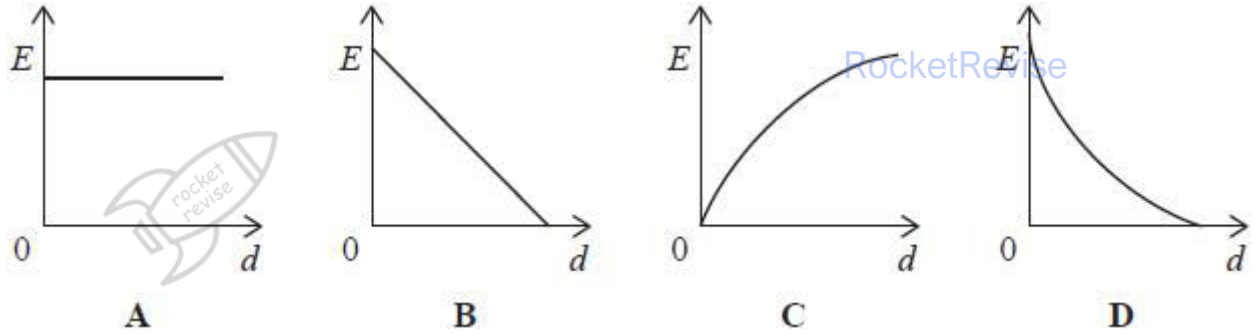
(Total for question = 1 mark)

Q28.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

A potential difference is applied across two parallel metal plates.

Which of the following graphs shows how the electric field strength *E* varies with distance *d* from the positive plate?



- ☒ **A**
- ☒ **B**
- ☒ **C**
- ☒ **D**

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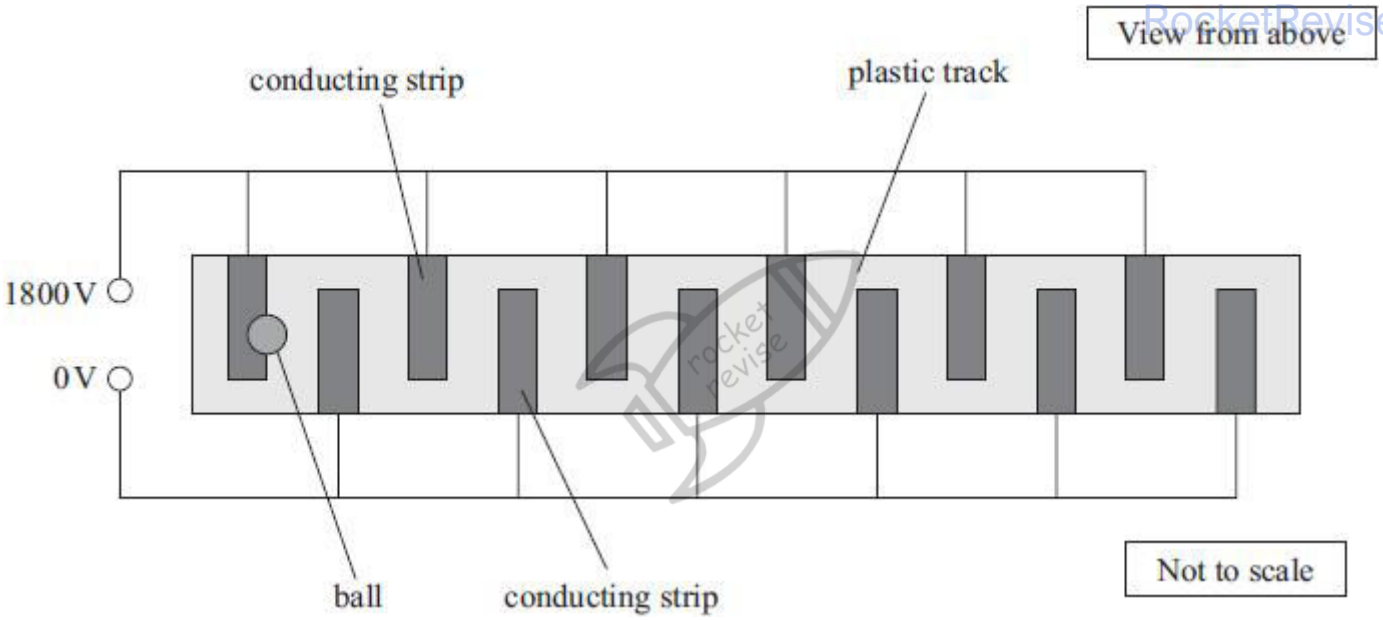
(Total for question = 1 mark)

Q29.

The diagram shows some equipment used by a teacher to demonstrate the action of electric fields.

A long white plastic track has strips of conducting material attached to it. The strips on one side of the track are connected to a potential of 1800 V. The strips on the other side are connected to a potential of 0 V.

A ball with a conducting surface can be accelerated along the track.



(a) The teacher places the ball on the first strip. The teacher switches on the power supply and the ball accelerates.

The size of the gap between the strips is 2.0 cm.

(i) The ball accelerates uniformly across the gap between the first strip and the second strip in 0.22 s.
Show that the acceleration of the ball is about 0.8 m s^{-2}.

(2)

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(ii) When placed on the first strip, the ball is charged to a potential of 1800 V.
Show that the charge on the ball is about 2×10^{-9} C.
radius of ball = 0.95 cm

(2)

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(iii) There is a force on the ball due to the uniform electric field between the strips.
Deduce whether this is the only force acting to accelerate the ball at about 0.8 m s^{-2}.
mass of ball = 0.18 g

(4)

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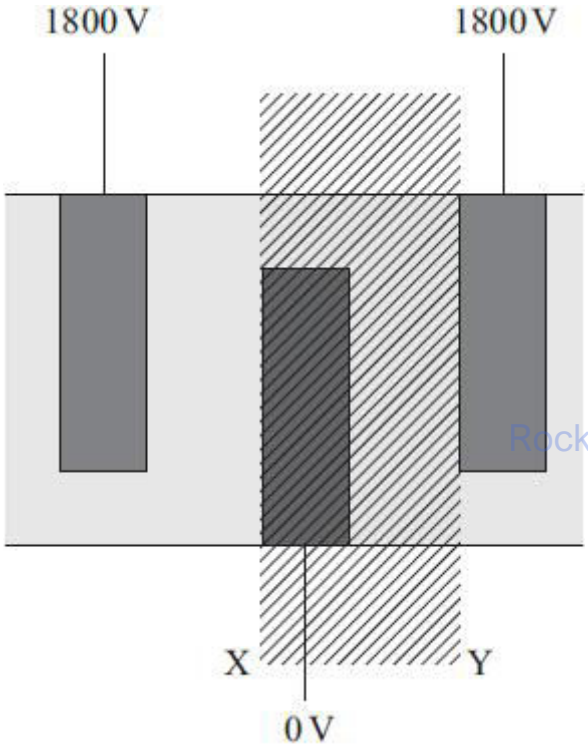
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(b) The equipment can be used as a model to demonstrate the acceleration of charged particles in a linac.

When placed on the first strip, the ball is charged to a potential of 1800 V. The ball then accelerates towards the next strip at 0 V, where it discharges.
The ball then moves at constant speed to the next 1800 V strip. The ball is charged again and accelerates to the next 0 V strip.
This pattern of movement continues until the ball reaches the end of the track.
The diagram shows part of the model.



(i) The shaded section of the diagram between X and Y corresponds to the drift tube in a linac.
Give **one** similarity and **two** differences in the way this section of the model works and the way a drift tube in a linac works.

(3)

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(ii) In the model, the lengths corresponding to drift tubes are constant.
Explain why the lengths of the drift tubes in a linac are not constant.

(3)

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(Total for question = 14 marks)

Q30.

The diagram represents a positive point charge.

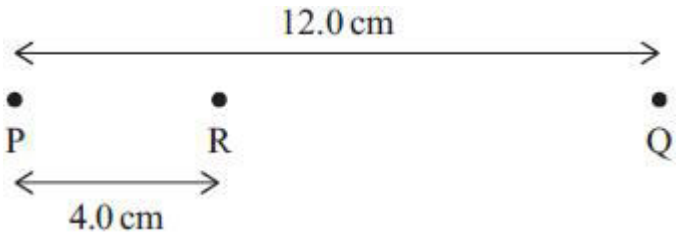


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(a) Draw field lines to show the electric field around the charge.

(3)

(b) The positive point charge is placed at P. A negative point charge is placed at Q, as shown below.



R is 4.0 cm from P, as shown.
Calculate the magnitude of the electric field strength at R.

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charge at P = +14 nC
charge at Q = −14 nC

(3)

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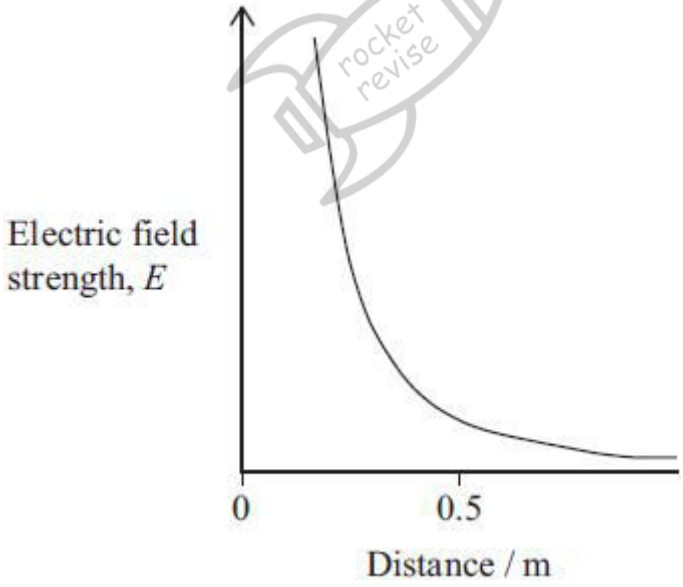
Magnitude of electric field strength = ...

(Total for question = 6 marks)

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Q31.

The graph shows how electric field strength varies with distance from a charged conducting sphere.



Which of the following gives the electric potential at a distance of 0.5 m from the sphere?

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- ☒ **A** The area under the graph from 0 to 0.5 m.
- ☒ **B** The area under the graph from 0.5 m to infinity.
- ☒ **C** The gradient of the tangent to the graph at 0.5 m.
- ☒ **D** The value of *E* at 0.5 m divided by 0.5 m.

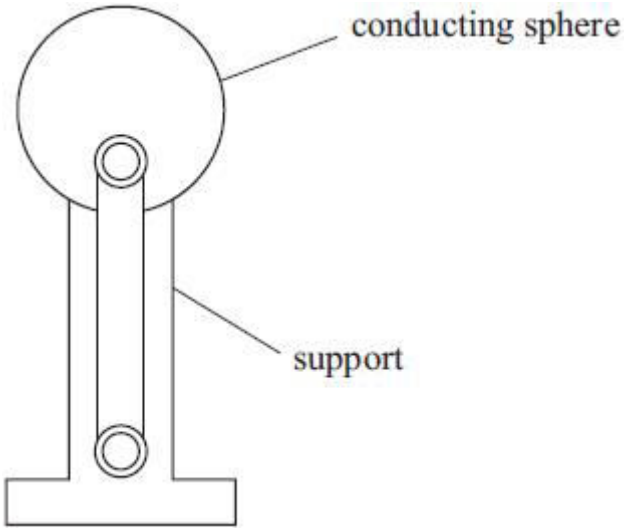
(Total for question = 1 mark)

Q32.

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The diagram shows a Van de Graaff generator. This is a device that produces a large potential difference by charging a conducting sphere.



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The conducting sphere of a Van de Graaff generator has a radius of 22.5 cm.

(a) The electric potential produced at the surface of the conducting sphere is 1.75×10^4 V.

Show that the charge on the sphere is about 4.4×10^{-7} C.

(2)

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(b) Neon bulbs light when placed in a region of strong electric field.

A neon bulb is moved towards the charged sphere of the Van de Graaff generator. The neon bulb lights when the distance between the neon bulb and the centre of the charged sphere is 32.5 cm. Determine the electric field strength at this position of the neon bulb.

(2)

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Electric field strength = ...

(Total for question = 4 marks)

Q33.

Answer the question with a cross in the box you think is correct ☒. If you change your mind

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about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Two identical charged particles were placed in a vacuum 0.020 m apart. The charge on each particle was $+8.0 \times 10^{-6}$ C.

Which of the following gives the magnitude of the force, in newtons, on each particle?

☒ **A** $\dfrac{8.99 \times 10^{9} \times (8.0 \times 10^{-6})}{0.020}$

☒ **B** $\dfrac{8.99 \times 10^{9} \times (8.0 \times 10^{-6})^{2}}{0.020^{2}}$

☒ **C** $\dfrac{8.99 \times 10^{9} \times (8.0 \times 10^{-6})^{2}}{0.020}$

☒ **D** $\dfrac{8.99 \times 10^{9} \times (8.0 \times 10^{-6})}{0.020^{2}}$

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(Total for question = 1 mark)

Q34.

In the early 20th century, Millikan carried out an experiment to determine the magnitude of *e*, the electronic charge. Charged oil droplets were introduced into a region of electric field produced between two parallel plates.

A potential difference was connected across the two plates.



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(a) Add to the diagram to show the field lines between the plates.

(2)

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(b) A charged oil drop entered the region of electric field between the plates. The plates were 1.5 cm apart. A potential difference of 85 V was applied between the plates and the charged oil drop was brought to rest.

Deduce whether the charge on the oil drop was a whole number multiple of the electronic charge.
volume of oil drop = 5.0×10^{-19} m^3
density of oil = 920 kg m^{-3}

(5)

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(Total for question = 7 marks)

Q35.

Answer the question with a cross in the box you think is correct (☒). If you change your mind about an answer, put a line through the box (☒) and then mark your new answer with a cross (☒).

A proton is in a vacuum. At a distance *r* from the proton, the electric field strength due to the proton is 0.77 N C^{-1}.

Which of the following gives the distance *r*, in m?

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☒ **A** $\sqrt{\dfrac{1.6 \times 10^{-19}}{4\pi \times 8.85 \times 10^{-12} \times 0.77}}$

☒ **B** $\sqrt{\dfrac{(1.6 \times 10^{-19})^2}{4\pi \times 8.85 \times 10^{-12} \times 0.77}}$

☒ **C** $\dfrac{1.6 \times 10^{-19}}{4\pi \times 8.85 \times 10^{-12} \times 0.77}$

☒ **D** $\dfrac{(1.6 \times 10^{-19})^2}{4\pi \times 8.85 \times 10^{-12} \times 0.77}$

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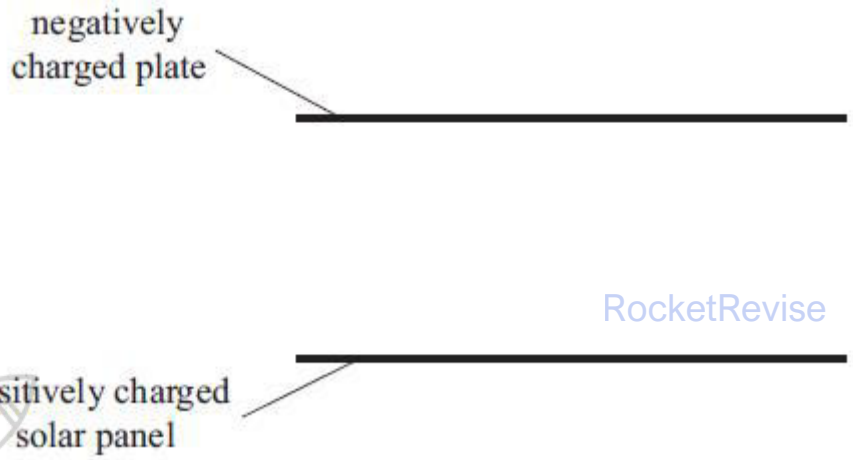
(Total for question = 1 mark)

Q36.

In desert areas of California, solar panels use sunlight to generate electrical power.
If sand collects on the solar panels, their efficiency is reduced.

To remove sand, a positive electric charge is given to the sand particles and a negatively charged plate is moved across the panel.

(a) The diagram shows a negatively charged plate above a positively charged solar panel.



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Add to the diagram to show the electric field lines between the plate and the solar panel.

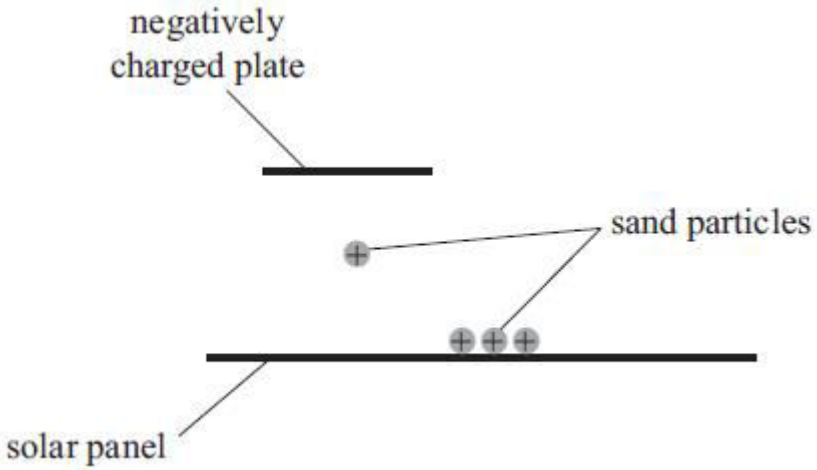
(2)

(b) A positive electric charge is given to the sand particles when the solar panel is charged.

When the negatively charged plate is moved across the solar panel, the sand particles move off the panel, as shown below.

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(i) Explain why a sand particle moves as shown.

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(2)

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(ii) Show that the electric field strength between the negatively charged plate and the solar panel is about 8.1 × 105 V m^{-1}.
distance between plate and solar panel = 1.50 cm
potential difference between plate and solar panel = 12.2 kV

(2)

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(iii) The maximum positive charge that can be given to the sand particles when the solar panel is charged is 4.0 × 10^{-12} C.
The table below shows the diameters of some sand particles.

Diameter of sand particle / μm	310	590	720	970

Only some of these particles could be lifted from the solar panel when using an electric field strength of 8.1 × 10^{5} V m^{-1}.
Deduce the largest diameter of sand particles given in this table that could be lifted from the solar panel.
density of sand = 2600 kg m^{-3}

(6)

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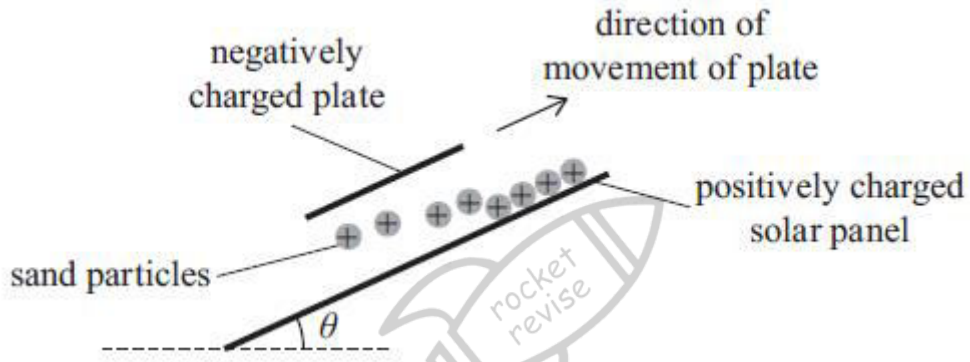
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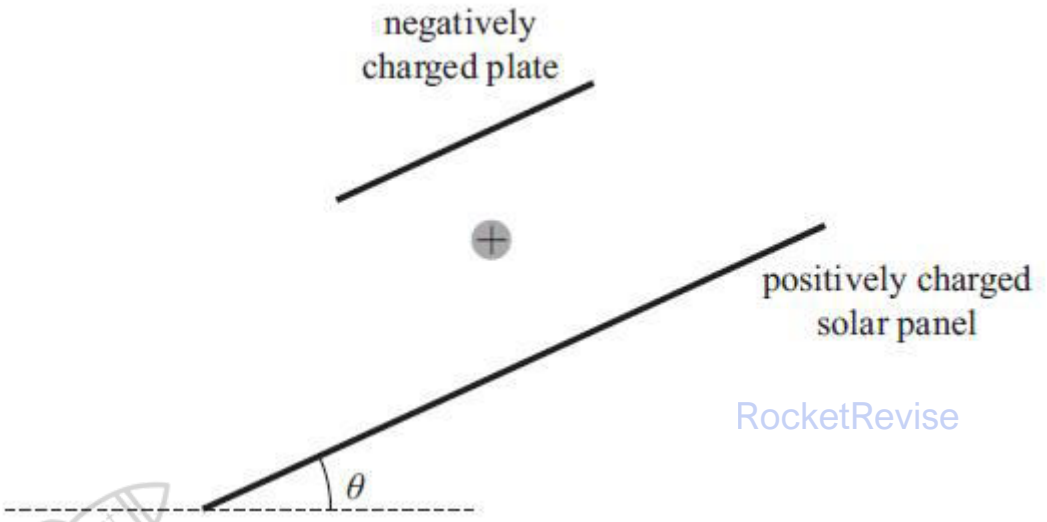
(c) The solar panels are usually inclined at an angle θ to the horizontal. The plate moves parallel to the solar panel, as shown.

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(i) Add to the diagram below to show the forces acting on a charged sand particle.

(2)



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(ii) Explain how inclining the solar panel at an angle θ would improve the removal of sand particles from the solar panel.

(2)

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